MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)

Fourth Semester Examination

(2024 Admission Onwards)

MG4VACMAT200 Business Mathematics

Duration: $1\frac{1}{2}$ hours Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall attainability.

Part A

Answer any five questions. Each question carries 2 marks.

1. Calculate the range of the following data that shows the marks in Mathematics obtained by 30 students in a class: [U][1]

37	38	40	36	38	37	36	40	50	47	41	46	31	33	48
37	32	26	46	32	26	33	48	37	42	16	41	36	38	48

2. Name any two types of bar graphs. [U][1]

3. Find the slope and y-intercept of the straight line $y = 2x - 4$. [U][2]

4. Determine the nature of solutions for the simultaneous equations: [U][2]

$$3x + 2y = 8$$
$$6x + 4y = 16.$$

5. Judy has an income of $Rs.10$ which she spends on fish and chips, the prices of which are $Rs.2$ and $Rs.3$ respectively. Write the inequality that represents the budget constraint. [U][3]

6. Define identity matrix. [U][4]

7. If $A = \begin{bmatrix} 7 & 4 \\ 6 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 9 \\ 8 & 2 \end{bmatrix}$, find the sum $A + B$. [U][4]

8. A company with four retail stores has 35 TVs (t), 60 stereos (s), 55 VCRs (v) and 45 camcorders (c) in store 1; $80\,t$, $65\,s$, $50\,v$ and $38\,c$ in store 2; $29\,t$, $36\,s$, $24\,v$ and $32\,c$ in store 3; $62\,t$, $49\,s$, $54\,v$ and $33\,c$ in store 4. Express present inventory in matrix form. [A][4]

Part B

Answer any four questions. Each question carries 5 marks.

9. The monthly productions of cars by a company for the first six months of a particular year are given below:
   January - 150, February - 200, March - 180, April - 230, May - 210, June - 140.
   Represent the production figures by a Line Chart. [*U*][1]

10. Construct a frequency polygon from the following data: [*U*][1]

Weight in Kg.	**No. of Students**
30-40	400
40-50	500
50-60	700
60-70	300
70-80	100
Total	**2000**

11. Suppose the demand and supply schedules for milk are:
$$q = 15 - 3\,p$$
$$q = 2\,p - 5$$
respectively. Find the equilibrium price and quantity of milk. [*U*][2]

12. A company produces three types of products (A, B, C) using two precious metals: Platinum and Gold. Due to market shortages, daily supplies of these metals are limited. The relevant data with respect to supply, requirements and profits are summarised in the table below:

Product	**Platinum required (g/unit)**	**Gold required (g/unit)**	**Profit/unit (Rs)**
A	2	3	500
B	4	2	600
C	6	4	1200

   Daily allotment of platinum and gold are 160 gm and 120 gm respectively.
   Formulate a Linear Programming Problem to determine the number of units of each product to be produced in order to maximize the profit. [*A*][3]

13. Check whether the matrices $A = \begin{bmatrix} 4 & 1 \\ 6 & 9 \\ 7 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 8 & 6 \\ 4 & 3 & 1 \end{bmatrix}$ are conformable for multiplication in the given order. If conformable, find the product AB. [*U*][4]

14. Find the value of the determinant of the matrix $A = \begin{bmatrix} 7 & 1 & 4 \\ 5 & 2 & 9 \\ 8 & 6 & 3 \end{bmatrix}$. [*U*][4]

Part C

Answer any two questions. Each question carries 10 marks.

15. Draw a Pie Chart for the following data relating to the production cost of manufacture: Cost of Materials - $Rs.38,400$; Cost of Labour - $Rs.30,720$; Direct Expenses of Manufacture - $Rs.11,520$; Factory Overhead Expenses - $Rs.15,360$. $[U][1]$

16. Solve the following LPP graphically: $[A][3]$

$$\begin{gathered}\text{Maximise } Z = 10\,x_1 + 15\,x_2 \quad \text{subject to} \\ 2\,x_1 + x_2 \leq 26 \\ 2\,x_1 + 4\,x_2 \leq 56 \\ x_1 - x_2 \geq -5 \\ x_1, x_2 \geq 0\end{gathered}$$

17. Show that the matrix $A = \begin{bmatrix} 4 & 2 & 7 \\ 3 & 1 & 8 \\ 5 & 3 & 1 \end{bmatrix}$ is non-singular and find its inverse A^{-1}. $[U][4]$

18. In an economy, the IS and LM equations are given by

$$\begin{aligned} 0.3\,Y + 100\,i &= 252 \\ 0.25\,Y - 200\,i &= 176 \end{aligned}$$

respectively, where Y is the national income and i is the rate of interest.

(a) Write the above system of equations in matrix form.

(b) Solve the system of equations to find the equilibrium values of the national income (Y) and the rate of interest (i). $[A][4]$